



NAME : BISMA FATIMA

CLASS : CS A

ROLL NO: F24605036

COURSE : COMPUTER NETWORK

COURSE CODE : CS-271

LECTURER : SIR NAVEED YOUSAF

LAB TASK 01

Introduction to Network Devices and Basic Commands

1. Network Configuration

Step 1. Connect to the Internet.

Establish and verify connectivity to the Internet. This step ensures the computer has an IP address.

Step 2. Gather TCP/IP configuration information.

- Use the Start menu to open the command prompt (Start>Programs>Accessories>Command Prompt or Start>Programs>Command Prompt).
- Type **ipconfig** and press Enter key. The spelling of the ipconfig is critical, but the case is not.
- The screen shows the IP address, subnet mask and the default gateway. The IP address and the default gateway should be in the same network or subnet; otherwise this host wouldn't be able to communicate outside the network

OUTPUT

```
C:\Users\M A D I N A>ipconfig
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix  . : mshome.net
Link-local IPv6 Address . . . . . : fe80::f2a:7349:1487:3930%10
IPv4 Address. . . . . : 192.168.137.63
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.137.1
```

Step 3. Record the following TCP/IP information for this computer.

- IP address:** fe80::f2a:7249:1487:3930%10
- Subnet mask:** 255.255.255.0
- Default gateway:** 192.168.137.1

Step 4. Compare this computer's TCP/IP configuration to that of others on the LAN. If this computer is on a LAN, compare the information of several machines (Hosts).

- Are there any similarities?**

Yes, several similarities usually exist:

- All devices will have **IP addresses within the same subnet**.
- They'll share the **same subnet mask**.
- The **default gateway** is typically identical across all hosts.
- DNS server settings may also be the same if configured centrally.

b) What is similar about the IP addresses?

- The network portion of the IP address is the same for all devices.
 - **For example**, in a subnet like 192.168.1.x, all devices might have IPs like 192.168.1.2, 192.168.1.3, etc.
- This similarity ensures that devices can communicate locally without routing through external networks.

c) What is similar about the default gateway?

The **default gateway** is usually the **IP address of the router** or central device that connects the LAN to external networks (like the internet).

All devices use this same gateway to send traffic outside the LAN.

- Example: 192.168.1.1 is a common default gateway in home and office networks.

d) Record a couple of the IP addresses (of your nearby hosts)

1. 192.168.137.242

2. 192.168.137.1

3. 192.168.137.68

Step 5. Check additional TCP/IP configuration information.

To see more information, type **ipconfig/all** and press Enter key. The figure shows the detailed IP configuration of this computer on the screen.

OUTPUT

```
C:\Users\M A D I N A>ipconfig/all
```

Windows IP Configuration

```

Host Name . . . . . : DESKTOP-M5NRCKD
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mshome.net

```

Wireless LAN adapter Wi-Fi:

```

Connection-specific DNS Suffix . : mshome.net
Description . . . . . : Intel(R) Dual Band Wireless-AC 8265
Physical Address. . . . . : 74-E5-F9-FF-A5-2D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::f2a:7349:1487:3930%10(Preferred)
IPv4 Address. . . . . : 192.168.137.63(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, September 18, 2025 1:37:35 AM
Lease Expires . . . . . : Thursday, September 25, 2025 2:19:16 AM
Default Gateway . . . . . : 192.168.137.1
DHCP Server . . . . . : 192.168.137.1
DHCPv6 IAID . . . . . : 158655993
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-A7-53-39-8C-EC-4B-FD-50-E9
DNS Servers . . . . . : 192.168.137.1
NetBIOS over Tcpip. . . . . : Enabled

```

- a) You should see the following information: the host name (computer name), the Physical address of this machine, IP address, subnet Mask, Default Gateway and DNS Servers.

1. the host name (computer name): DESKTOP-M5NRCKD
2. the Physical address : 74-E5-FF-A5-WD
3. IP address: 192.168.137.63
4. subnet Mask: 255.255.255.0
5. Default Gateway: 192.168.137.1
6. DNS Servers: 192.168.137.1

- b) In the LAN, compare your result with a few nearby computers. What similarities do you see in the physical (MAC) address

MAC addresses have a consistent format and often share the same manufacturer prefix (OUI) if devices are from the same vendor. Each MAC address is still unique to avoid conflicts on the LAN.

- c) Write down the computer's host name: DESKTOP-M5NRCKD
- d) Write down the host names of a couple of other computer:

1. AYESHA

2. DESKTOP-196N200
3. Saif_PROBOOK
4. DESKTOP-TM6DVG3

1. Using PING, TRACERT , PATHPING from a Workstation

Sept 1. Establish and verify connectivity to the Internet. This step ensures that the computer has an IP address.

Step 2. Open the Command prompt. Ping the IP address of another computer.

- a. In the window, type **ping**, a space, and the **IP address of a computer** recorded above.

```
C:\Users\M A D I N A> ping 192.168.137.63

Pinging 192.168.137.63 with 32 bytes of data:
Reply from 192.168.137.63: bytes=32 time<1ms TTL=128
Reply from 192.168.137.63: bytes=32 time<1ms TTL=128
Reply from 192.168.137.63: bytes=32 time<1ms TTL=128
Reply from 192.168.137.63: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.137.63:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\M A D I N A>
```

Ping uses the Internet Control Message Protocol(ICMP) echo-request and echo-reply feature to test physical connectivity. Because ping reports on four attempts, it gives an indication the reliability of the connection. Look over the result and verify that the ping was successful. Was the ping successful? If not, report to the instructor.

- a) Ask the IP address of the nearby computers and ping. Note the result.
 1. **192.168.137.242**
 2. **192.168.137.184**
 3. **192.168.137.288**
- b) Ping the IP address of Default gateway and DNS servers. Was the result successful?

```
C:\Users\M A D I N A>ping 192.168.137.1

Pinging 192.168.137.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.137.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\M A D I N A>
```

No its no working for gateway and DNS

c) Ping the computer's loopback IP address. Type the following command:

ping 127.0.0.1

The address 127.0.0.1 is reserved for loopback testing. If the ping is successful, then TCP/IP is properly installed and functioning on this computer.

```
C:\Users\M A D I N A>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

d. Was the ping successful for d._

```
C:\Users\M A D I N A>ping d
Ping request could not find host d. Please check the name and try again.
```

No its is not successful

e. Ping the hostname of the computer that you recorded in previous task.

```
C:\Users\M A D I N A>ping DESKTOP-M5NRCKD

Pinging DESKTOP-M5NRCKD [fe80::e923:6460:204b:5404%4] with 32 bytes of data:
Reply from fe80::e923:6460:204b:5404%4: time<1ms
Reply from fe80::e923:6460:204b:5404%4: time<1ms
Reply from fe80::e923:6460:204b:5404%4: time<1ms
Reply from fe80::e923:6460:204b:5404%4: time<1ms

Ping statistics for fe80::e923:6460:204b:5404%4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

f. Ping the Microsoft website(www.microsoft.com) and record result.

```
C:\Users\M A D I N A>ping www.microsoft.com

Pinging e13678.dscb.akamaiedge.net [184.25.221.172] with 32 bytes of data:
Reply from 184.25.221.172: bytes=32 time=103ms TTL=48
Reply from 184.25.221.172: bytes=32 time=104ms TTL=48
Reply from 184.25.221.172: bytes=32 time=103ms TTL=48
Reply from 184.25.221.172: bytes=32 time=101ms TTL=48

Ping statistics for 184.25.221.172:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 101ms, Maximum = 104ms, Average = 102ms
```

Step 3. Trace the route to the NUTECH university website: type

tracert www.nutech.edu.pk and press Enter key.

The result shows the complete route to the site and the number of hops in path.

```

C:\Users\M A D I N A>tracert www.nutech.edu.pk

Tracing route to www.nutech.edu.pk [203.124.43.228]
over a maximum of 30 hops:

  1    4 ms    *        4 ms    CL01-L3-PC21.mshome.net [192.168.137.1]
  2    *      *        *        Request timed out.
  3    3 ms    2 ms     1 ms    10.5.1.1
  4   12 ms    1 ms    41 ms    10.1.0.10
  5   12 ms    1 ms    1 ms    mbl-99-54-233.dsl.net.pk [203.99.54.233]
  6    2 ms    2 ms    3 ms    172.16.33.57
  7   15 ms    2 ms    3 ms    172.31.5.29
  8    5 ms    2 ms    2 ms    172.31.5.206
  9    2 ms    2 ms    2 ms    172.31.5.170
 10    4 ms    3 ms    5 ms    103.213.108.87
 11    *      *        *        Request timed out.
 12    *      6 ms   10 ms   host201003.comsatshosting.com [203.124.43.228]

Trace complete.

C:\Users\M A D I N A>

```

Step 4. Close the window.

Lab Task 02

Explore 10 cmd networking related commands.

1. **ping** - Tests connectivity to a remote host using ICMP packets.

```

C:\Users\M A D I N A>ping google.com

Pinging google.com [142.250.202.174] with 32 bytes of data:
Reply from 142.250.202.174: bytes=32 time=43ms TTL=111
Reply from 142.250.202.174: bytes=32 time=47ms TTL=111
Reply from 142.250.202.174: bytes=32 time=46ms TTL=111
Reply from 142.250.202.174: bytes=32 time=160ms TTL=111

Ping statistics for 142.250.202.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 160ms, Average = 74ms

```


2. **ipconfig** - Displays IP address, subnet mask, and gateway for network adapters.

Wireless LAN adapter Wi-Fi:

```

Connection-specific DNS Suffix . : mshome.net
Link-local IPv6 Address . . . . . : fe80::f2a:7349:1487:3930%10
IPv4 Address. . . . . : 192.168.137.63
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.137.1

```

3. **tracert** - Traces the route packets take to a destination host.

Tracing route to google.com [142.250.202.174]
over a maximum of 30 hops:

```

 1    80 ms    *          7 ms  CL01-L3-PC21.mshome.net [192.168.137.1]
 2      *      *          *      Request timed out.
 3     1 ms    1 ms     1 ms  10.5.1.1
 4     4 ms    4 ms     9 ms  10.1.0.10
 5     2 ms    5 ms     2 ms  mbl-99-54-233.dsl.net.pk [203.99.54.233]
 6     5 ms   19 ms    26 ms  172.16.33.57
 7     3 ms   18 ms     4 ms  172.31.5.29
 8     3 ms    2 ms     4 ms  172.31.5.206
 9     4 ms    7 ms     6 ms  172.31.5.170
10     3 ms    3 ms     2 ms  110.93.202.134
11    27 ms   27 ms    32 ms  110.93.253.126
12    35 ms   30 ms    23 ms  110.93.252.246
13    72 ms   35 ms    35 ms  110.93.192.109
14    81 ms   43 ms    37 ms  192.178.252.233
15    35 ms   34 ms    37 ms  192.178.98.246
16    66 ms   36 ms    50 ms  142.251.77.153
17    41 ms   37 ms    46 ms  192.178.105.67
18    37 ms   36 ms    38 ms  192.178.96.7
19    74 ms   39 ms    42 ms  lcmcta-ae-in-f14.1e100.net [142.250.202.174]

```

4. **nslookup** - Queries DNS servers to resolve domain names to IP addresses.

```

C:\Users\M A D I N A>nslookup google.com
Server: CL01-L3-PC21.mshome.net
Address: 192.168.137.1

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4018:810::200e
           142.250.186.238

```

5. netstat - Shows active network connections and listening ports.

```

C:\Users\M A D I N A>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    192.168.137.63:60264    4.213.25.240:https      ESTABLISHED
TCP    192.168.137.63:60268    4.213.25.240:https      ESTABLISHED
TCP    192.168.137.63:60424    mbl-165-249-97:https    CLOSE_WAIT
TCP    192.168.137.63:60425    110.93.229.162:https    CLOSE_WAIT
TCP    192.168.137.63:60426    whatsapp-cdn-shv-01-sin2:https CLOSE_WAIT
TCP    192.168.137.63:60427    whatsapp-cdn-shv-01-mct1:https CLOSE_WAIT
TCP    192.168.137.63:60428    ntl-50-113-162:https    CLOSE_WAIT
TCP    192.168.137.63:60429    whatsapp-cdn-shv-01-mct1:https CLOSE_WAIT
TCP    192.168.137.63:60456    a2-16-158-42:https      ESTABLISHED
TCP    192.168.137.63:60460    wr-in-f188:5228         ESTABLISHED
TCP    192.168.137.63:60461    57.155.141.115:https    ESTABLISHED
TCP    192.168.137.63:60463    13.69.239.68:https      ESTABLISHED
TCP    192.168.137.63:60466    104.208.16.89:https     ESTABLISHED
TCP    192.168.137.63:60470    173:https               ESTABLISHED
TCP    192.168.137.63:60472    249:https               ESTABLISHED
TCP    192.168.137.63:60476    vps-db726223:10343      ESTABLISHED

```

6. arp - Displays and modifies the ARP cache (IP-to-MAC address mapping).

```
C:\Users\M A D I N A>arp -a
```

```
Interface: 192.168.147.1 --- 0x4
```

Internet Address	Physical Address	Type
192.168.147.254	00-50-56-f8-86-dd	dynamic
192.168.147.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.137.63 --- 0xa
```

Internet Address	Physical Address	Type
192.168.137.1	50-3e-aa-60-b5-2f	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.146.1 --- 0x12
```

Internet Address	Physical Address	Type
192.168.146.254	00-50-56-fb-19-5d	dynamic
192.168.146.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

7. getmac - Shows the MAC address of your network adapters.

```
C:\Users\M A D I N A>getmac
```

Physical Address	Transport Name
8C-EC-4B-FD-50-E9	Media disconnected
74-E5-F9-FF-A5-2D	\Device\Tcpip_{6744163E-58A7-4C32-92E8-80402F4840C8}
74-E5-F9-FF-A5-31	Media disconnected
00-50-56-C0-00-01	\Device\Tcpip_{DC38BE96-A18A-4489-9552-48BA987F1477}
00-50-56-C0-00-08	\Device\Tcpip_{245D7D91-3C62-4090-9A60-73BCCDCCEDB6}

8. hostname - Displays the name of the current computer.

```
C:\Users\M A D I N A>hostname
DESKTOP-M5NRCKD
```

```
C:\Users\M A D I N A>netsh interface show interface
```

Admin State	State	Type	Interface Name
Enabled	Connected	Dedicated	VMware Network Adapter VMnet1
Enabled	Connected	Dedicated	VMware Network Adapter VMnet8
Enabled	Disconnected	Dedicated	Ethernet
Enabled	Connected	Dedicated	Wi-Fi

9. netsh - Configures network interfaces, firewall, and other settings.

```
C:\Users\M A D I N A>netsh wlan show interfaces
```

There is 1 interface on the system:

```

Name                : Wi-Fi
Description          : Intel(R) Dual Band Wireless-AC 8265
GUID                 : 6744163e-58a7-4c32-92e8-80402f4840c8
Physical address     : 74:e5:f9:ff:a5:2d
State                : connected
SSID                 : Ayan
BSSID                : b0:19:21:ee:c1:aa
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Profile
Channel              : 3
Receive rate (Mbps)  : 300
Transmit rate (Mbps) : 300
Signal               : 91%
Profile              : Ayan

Hosted network status : Not available

```

```
C:\Users\M A D I N A>_
```

10. pathping - Combines ping and tracert to show packet loss at each hop.

```
C:\Users\M A D I N A>pathping google.com

Tracing route to google.com [142.250.202.174]
over a maximum of 30 hops:
  0  DESKTOP-M5NRCKD.mshome.net [192.168.137.63]
  1  CL01-L3-PC21.mshome.net [192.168.137.1]
  2  * * *
Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
    0          Lost/Sent = Pct  Lost/Sent = Pct  DESKTOP-M5NRCKD.mshome.net [192.168.137.63]
    |
  1  ---      100/ 100 =100%    0/ 100 = 0%     CL01-L3-PC21.mshome.net [192.168.137.1]

Trace complete.
```

Lab Task 03

Create your cisco id (<https://id.cisco.com/signin/register>)

Create Account

* indicates required field

Email *

Password *



First name *

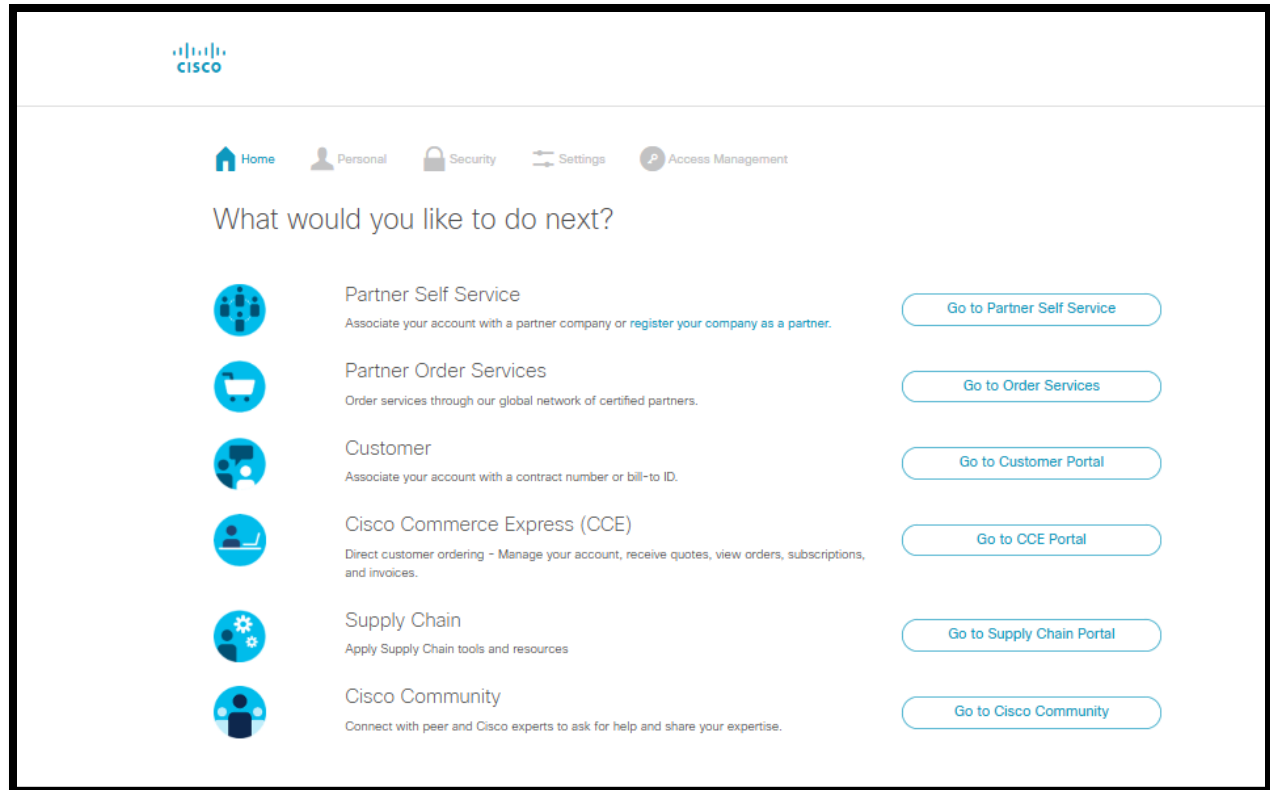
Last name *

Country or region *



By clicking Register, I confirm that I have read and agree to the [Cisco Online Privacy Statement](#) and the [Cisco Web Site Terms and Conditions](#).

Register



Lab Task 04

Study of following Network Devices in Detail

Repeater

- Operates at the physical layer of the OSI model.
- Regenerates and amplifies weak signals to extend network range.
- Used in both wired and wireless networks.
- Does not filter or route data—just boosts it.
- Ideal for long-distance communication where signal loss occurs.

Hub

- Connects multiple devices in a LAN and broadcasts data to all ports.
- Works at the physical layer and lacks intelligence.
- Cannot filter data or identify destinations.
- Creates a single collision domain, reducing efficiency.

- Comes in active, passive, and intelligent types.

Switch

- Operates at the data link layer (Layer 2) of the OSI model.
- Forwards data only to the intended device using MAC addresses.
- Supports full-duplex communication and reduces collisions.
- More efficient than hubs, with multiple collision domains.
- Can be managed or unmanaged depending on network needs.

Bridge

- Connects and filters traffic between two LAN segments.
- Works at the data link layer and uses MAC addresses.
- Reduces network traffic by forwarding only necessary data.
- Helps segment networks for better performance.
- Can be transparent, source-routing, or translational.

Router

- Operates at the network layer (Layer 3) of the OSI model.
- Routes data between different networks using IP addresses.
- Maintains routing tables to determine best paths.
- Connects LANs to WANs and the internet.
- Supports NAT, firewall, and VPN functionalities.

Gateway

- Connects networks using different protocols and architectures.
- Acts as an entry/exit point for data between networks.
- Performs protocol conversion for compatibility.
- Can be hardware, software, or a combination.

- Often integrated with routers in modern setups